7th June 2026

Andrej Karpathy.

Decoding – what's behind the chatGpt?
How chatGpt is built?

Stages →

pre-training ⟶ step1 :- download & preprocess the internet.

⇓

FineWeb : by Huggingface
↓
44Tb disk space, 15-trillion tokens.

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URL filtering → Text Extraction → language filtering → Gopher filtering
↓
PII Removal. ← custom filters ← C4 filters ← MinHash dedup

① Finding the raw data
↳ Common Crawl.

② URL filtering using a blocklist

③ Text Extraction using fastext language classifier.

⇓

Removing HTML tags, CSS code and just keeping the text

⇓

Used only webpage with 65% of English text

step-2:- tokenization. @codingdidi

e.g.→ consider 200 pages text

Neural Network cannot Understand text we want One dimensional sequence.

So, convert text into bits (8 digit) ∴ combination of 0's and 1's.

Then bits to bytes ∴ 256 possible Combination

⇓

0 to 255 ∴ Unique ID's or Unique 255 emoji's

M	T	W	T	F	S	S
			1	2	3	4
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26	27	28	29	30	31	

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This is generally done by running what's called
"The Bite-pair encoding"

Algorithm @codingdidi

Target is to - shrink the data as much as possible.

Use → Tiktokenizer.vercel.app
↓
cl100k_base model.

Tokenization :- raw text into sequence of symbols.

∴ Case sensitive
∴ Spaces forms New tokens.

→ 40,000 bits (2 possible tokens)
→ 5000 bytes (256 possible token)

→ 1300 GPT-4 tokens (100,277 possible tokens)

Step 3 :- Neural Network training

⇓

Computationally high.

⇒ consider a random set of tokens and then predict the Next token

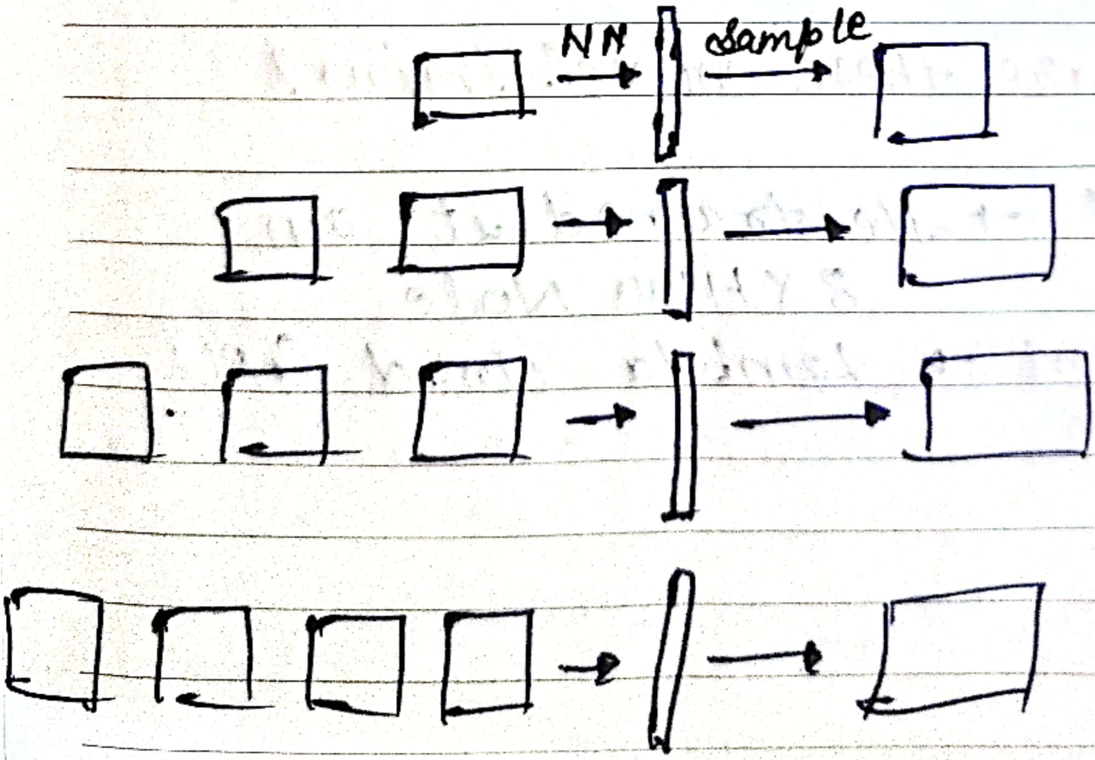
sequences of token – Known as Context

input × weight.

Use :- bbycroft.net/llm @codingdidi

→ nano-gpt

inference :- to generate data, just predict one token at a time.



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GPT-2 → 2019 by 'Open AI'

paper → Language Models are Unsupervised Multitask learners

Transformer neural Network.

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- → 1.6 billion parameters
- → maximum context length of 1024 tokens.
- → trained on about 100 billions tokens.

Use → Karpathy / llm.c → Github

How & where are these model trains?

↳ on cloud → He trained it on 8 X H100 Node

rent → Lambda cloud GPU

Biggest llm base models.

Cohere	GPT-4	Claude	Falcon
GPT-3	Llama	OpenAI	Gemini
Mistral AI	PaLM	Google	
Anthropic.	Nvidia.	BERT	
Meta	BLOOM	Microsoft	
llama 3.1	LaMDA		
xAI			

Base Model → token simulator or Internet text token simulator

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for llama 3 → Use → Herds of Model paper

To interact with base models use →

Hyperbolic.xyz

- stochastic Model
- predicting not Assisting
- Not Useful for user.

- Regurgitation is in base model

base model can be utilised in practical application.

Use – few shot prompt – Method

because they have in Context learning ability

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e.g + Try translator (English to Korean)
or
AI asistant (By giving Gasb & Description)

POST-Training :-

conversation, script

→ Using tcpip packet —For→ token
↓
User & Assistant → token

we basically need tokens of conversation
∴ User & Assistant.

Use → tiktokenizer → Gpt-4o

conversation dataset → Training language model to follow instruction with human feedback → (Paper)
2203.02155

② huggingface – dataset – OpenAssitant – Oasst1

③ UltraChat → LLM to generate dataset

Hallucinations
⇓ @codingdidi
⇒ false, fabricated or misleading output

Tested on → inference-playground → Huggingface
↓
falcon-7b-instruct

How to Correct?

① Use model interrogation to discover model's Knowledge and programmatically augment its training dataset with Knowledge-based refusal.

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⑤ Allow models to search → search_start search_end

What is context window?

⇓

"it is working memory"

→ It dictates how much text, code or data — measured in tokens — can be analyzed or remember at any given time by model. @codingdidi

Vagune recollection Vs Working Memory.

Knowledge in the parameters == Vague recollection (∴ of something you read 1 month ago)

Knowledge in the tokens of the context window == Working memory.

Huggingface → dataset → allenai →
tulu-3-sft-olmo-2-mixture

→ Opensource model → LLM.

- Hard Coded Conversation
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- Knowledge of self →
- Model needs tokens to think
- Model can't count
- Models are not good with spelling.

- Number Comparison

sft model

exposition ⟺ pretraining
(bg knowledge)

worked problem ⟺
supervised finetuning
(problem + demonstrated solution for imitation

practice problem ⟺ reinforcement learning.
(prompt to practice, trial & error ~~until~~ you reach the correct answer)

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RL model

as LLM

Use → Deepseek_R1

Incentivizing Reasoning Capability in LLM via Reinforcement learning. @codingclub

Dataset → art of problem solving.

Difference between sft and RL Model.

SFT ⟷ RL

Supervised fine tuning Tuning ⟷ Reinforcement learning.

process of taking a base model and training it on a high-quality dataset of human written prompt-response

takes SFT-trained model & improves it by allowing the model to generate its own answers and receive

Use → together.ai

Angels fly because they take themselves lightly.

Use → lmarena.ai
→ https://buttondown.com/ainews.
X/ Twitter.

proprietary model :→ on respective website of LLM providers.

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Open weights model :→ an inference provider e.g. Together AI.

Run them locally! LMStudio

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